The Effect of Inflation on GDP Growth in Czechia: An Econometric Analysis

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1. Introduction

Inflation and economic growth are among the most important macroeconomic indicators influencing the stability and development of modern economies. High inflation may reduce purchasing power, increase uncertainty, and negatively affect investment and consumption decisions, which can subsequently influence economic growth. According to Fischer (1993), macroeconomic instability associated with inflation may weaken economic performance and reduce long-run growth. Similarly, Barro (1995) finds evidence that higher inflation is associated with lower real GDP growth and lower investment activity.

The relationship between inflation and GDP growth is very complex and has been widely discussed in economic literature. While moderate inflation may accompany economic expansion, persistently high inflation may negatively affect long-term macroeconomic stability.

The aim of this paper is to examine the relationship between inflation and GDP growth in Czech Republic using econometric time-series model. The analysis is based on annual data covering the period of 25 years, from 2000 to 2024 obtained from the World Bank database. In addition to a simple linear regression model, a finite distributed lag (FDL) model is estimated in order to capture possible delayed effects of inflation on economic growth.

2. Theoretical Background

The relationship between inflation and economic growth has long been discussed in macroeconomic literature. Economic theory suggests that inflation can influence growth through several channels, including investment decisions, purchasing power, financial market uncertainty, and aggregate demand.

Fischer (1993) argues that macroeconomic instability associated with high inflation may reduce economic growth by lowering investment and productivity growth. His empirical analysis shows that economic growth is negatively associated with inflation and other indicators of macroeconomic instability.

Similarly, Barro (1995) examines data from approximately 100 countries and finds evidence of a negative relationship between inflation and economic performance. According to Barro, an increase in average inflation is associated with lower real GDP growth and reduced investment activity.

However, the effects of inflation on economic activity may not occur immediately. Firms and households often adjust gradually to changes in the price level, while monetary policy responses may affect the economy with delay. For this reason, it is appropriate to

examine not only the contemporaneous relationship between inflation and GDP growth, but also possible lagged effects through a distributed lag specification.

Based on the theoretical arguments and previous empirical findings, the following research hypothesis is formulated:

H1: Higher inflation negatively affects GDP growth in Czechia.

3. Data and Methodology

3.1 Data

The analysis uses annual macroeconomic data for Czechia covering the period 2000–2024. The data was obtained from the World Bank database and represent time-series data. The dependent variable is GDP growth measured as annual percentage growth of gross domestic product. Inflation is measured using the annual percentage change in consumer prices.

The final dataset contains 24 complete observations after the inclusion of the lagged inflation variable. Descriptive statistics for the analyzed variables are presented in Table 1.

Table 1: Descriptive statistics of analyzed variables

Variable	Mean	Median	Min	Max
Inflation	3.174	2.443	0.119	15.100
GDP Growth	2.360	2.772	-5.305	6.623
Inflation Lag	3.229	2.492	0.119	15.100

The average inflation rate during the analyzed period was approximately 3.17%, while average GDP growth reached 2.36%. Inflation showed substantial variability, particularly during the post 2021 inflationary period. GDP growth also fluctuated considerably due to major economic shocks such as the global financial crisis and the COVID-19 pandemic.

Figure 1: Inflation in Czechia

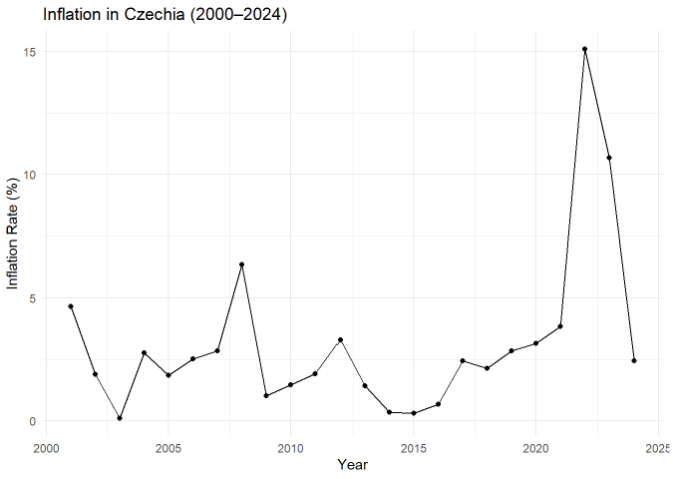


Figure 1 illustrates the development of inflation in Czechia between 2000 and 2024. Inflation remained relatively moderate with flacuations during most of the analyzed period, although a very sharp increase occurred after 2021, reaching a peak in 2022. This development likely reflects post-pandemic inflationary pressures and rising energy prices in Europe.

Figure 2: GDP Growth in Czechia (2000–2024)

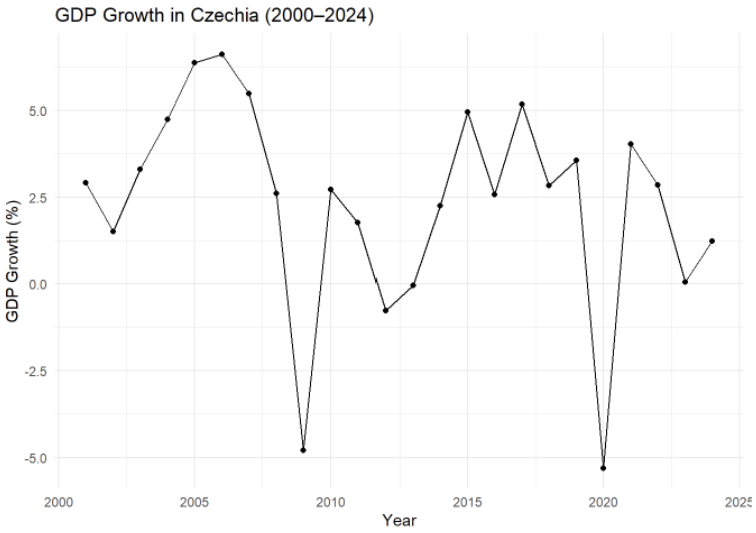
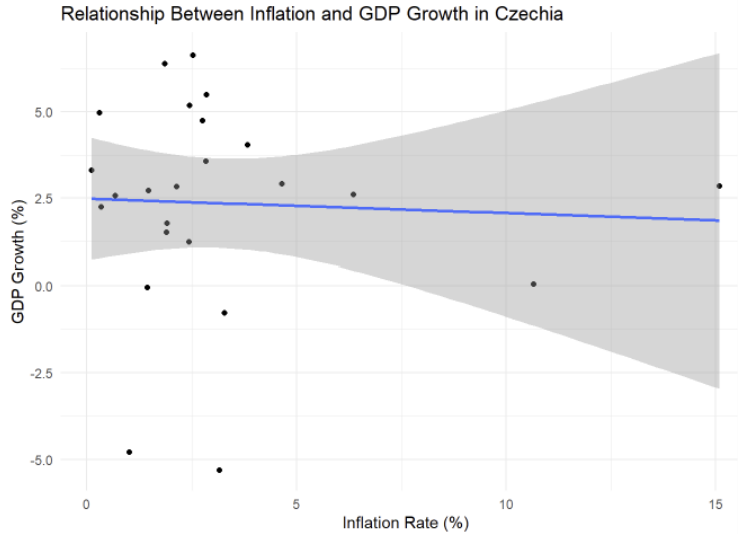


Figure 2 presents the evolution of GDP growth in Czechia during the analyzed period. GDP growth showed significant fluctuations, with considerable declines observed during the global financial crisis in 2009 and the COVID-19 pandemic in 2020, followed by periods of economic recovery.

3.2 Correlation Analysis

The correlation coefficient between inflation and GDP growth is −0.047, indicating a weak negative relationship between the variables.

Figure 3: Relationship between inflation and GDP growth in Czechia



The low correlation coefficient suggests that inflation alone has limited explanatory power for GDP growth in Czechia when only contemporaneous effects are considered. Figure 3 illustrates the relationship between inflation and GDP growth together with the fitted regression line. Although the regression line indicates a slightly negative relationship, the observations are widely dispersed, suggesting relatively weak explanatory power of simultaneous relationship of inflation and GDP growth.

3.3 Econometric Models

Two econometric models were estimated in this paper. First, a simple ordinary least squares (OLS) regression model was used to examine the relationship between inflation and GDP growth.

$$GDP_t = \beta_0 + \beta_1 Inflation_t + u_t$$

After that, a finite distributed lag (FDL) model was estimated in order to capture possible delayed effects of inflation on economic growth through the inclusion of lagged inflation.

$$GDP_t = \beta_0 + \beta_1 Inflation_t + \beta_2 Inflation_{\{t-1\}} + u_t$$

where (GDP_t) represents annual GDP growth in year (t), ($Inflation_t$) denotes the inflation rate, ($Inflation_{\{t-1\}}$) represents lagged inflation, and (u_t) is the error term.

The models were estimated using ordinary least squares in the R statistical software environment. To evaluate the reliability of the estimated models, diagnostic tests for heteroskedasticity and autocorrelation were performed using the Breusch–Pagan test and the Durbin–Watson test.

4. Results

Table 3 presents the estimated regression coefficients for the simple OLS model and the finite distributed lag (FDL) model.

The simple OLS regression model indicates a weak negative relationship between inflation and GDP growth. However, the coefficient of inflation is statistically insignificant (p = 0.827), suggesting that inflation does not significantly explain GDP growth in Czechia during the analyzed period.

The finite distributed lag model provides more informative results. While the contemporaneous inflation coefficient remains statistically insignificant, the lagged inflation variable exhibits a negative coefficient and is significant at the 10% significance level (p = 0.058). This finding suggests that inflation may negatively affect economic growth with delay rather than immediately.

Compared to the simple regression model, the FDL model also demonstrates improved explanatory power, as indicated by the increase in the coefficient of determination (R^2) from 0.002 to 0.162 and the lower residual standard error.

Table 3: Comparison of Regression Coefficients in OLS and FDL Models

Variable	OLS	FDL
Inflation	-0.041	0.138
	(0.188)	(0.198)
Inflation Lagged		-0.396*
		(0.198)
Constant	2.492***	3.204***
	(0.861)	(0.882)
R^2	**0.002**	**0.162**
Observations	**24**	**24**

*Notes: Standard errors are reported in parentheses. The dependent variable in all models is GDP growth. * p < 0.1, ** p < 0.05, *** p < 0.01.*

Moving to the tests of heteroskedasticity and autocorrelation, they did not indicate significant econometric problems in the estimated FDL model. The Breusch–Pagan test did not provide evidence of heteroskedasticity (p = 0.795), while the Durbin–Watson test did not indicate significant positive autocorrelation in the residuals (p = 0.155).

The estimated results are broadly consistent with the theoretical arguments presented by Fischer (1993) and Barro (1995), who both emphasize the negative effects of inflation on economic performance. Although the contemporaneous effect of inflation was statistically insignificant in the simple regression model, the finite distributed lag model suggests that inflation may negatively affect GDP growth with delay.

5. Discussion

The results suggest that inflation does not have a strong immediate effect on GDP growth in Czechia. However, the negative coefficient of lagged inflation indicates that inflationary pressures may influence economic activity with delay. This finding is economically plausible, as firms, households, and monetary authorities often adjust gradually to changes in the price level.

The obtained results are generally consistent with previous empirical studies. Fischer (1993) argues that macroeconomic instability associated with inflation may negatively affect growth through lower investment and productivity, while Barro (1995) finds evidence of a negative relationship between inflation and long-term economic performance.

Several limitations of the analysis should be acknowledged. First, the study is based on a relatively small annual dataset covering only one country. Second, GDP growth is influenced by many additional macroeconomic factors that are not included in the estimated models, such as unemployment, interest rates, fiscal policy, or multiple international economic conditions.

References

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AI Usage

AI tools were used for language correction, academic phrasing, and grammar improvement, as English is not my first language. It was also used to support the formulation and interpretation of selected econometric results and regression coefficients. The econometric analysis, data processing, coding, model estimation, visualisation and final evaluation of results were conducted without the use of AI.